Karan Kardam

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Education

University of Toronto — Sep 2024 – Present

Bachelor of Applied Science, Specializing in Computer Engineering, Minor in Artificial Intelligence

Work Experience

Software Engineer Intern — May 2026 – Aug 2026

Microsoft — *Redmond, Washington*

- Architected Perception Brain, an agentic RAG assistant answering ownership and onboarding questions with cited sources, on Microsoft Agent Framework (MAF) + MCP and served to 100+ daily users via GitHub Copilot CLI
- Automated ingestion of 1,000+ SharePoint and Teams docs through Microsoft Graph, using Azure OpenAI to extract people, workstreams, and ownership into a deduplicated knowledge graph with source-linked citations
- Engineered the RAG retrieval and answering layer on Azure AI Search, using hybrid BM25 keyword + vector embedding search fused by Reciprocal Rank Fusion so the agent cites every answer or refuses when unsure
- Designed a multi-agent system: an MCP server exposing wiki tools to Copilot CLI, a scheduled curator agent flagging contradictions and staleness under human-in-the-loop approval, and status roll-up queries
- Validated with a 100+ question eval set at 96% correctness on accepted answers and 100% citation recall

Machine Learning Researcher — Jan 2026 – Apr 2026

University of Toronto — *Toronto, Ontario*

- Built a two-stage multi-objective optimization framework (Differential Evolution + NSGA-II) for energy systems
- Deployed Dockerized PyTorch surrogate models behind a FastAPI REST service with Pydantic validation and batched inference, enabling scalable optimization workflows at 1.1s p95 latency
- Accelerated energy-grid scoring 10x with CuPy vectorization, NumPy batching, Ray parallelism, and LRU caching

Software Engineer Intern — May 2025 – Aug 2025

Foresters Financial — *Toronto, Ontario*

- Modernized a legacy COBOL-based tax processing system into a scalable C# .NET Core microservice architecture, reducing technical debt, improving maintainability, and enabling efficient tax record updates via modular services
- Deployed backend services using Docker, Azure Kubernetes Service (AKS), and Azure SQL Database, reducing deployment time by 70%, enabling autoscaling, and streamlining cloud infrastructure management
- Built a component-based React Vite and TypeScript frontend integrated with ASP.NET Core RESTful APIs using Entity Framework & SQL, cutting load time 40% by optimizing APIs and builds, and ensuring efficient data access
- Implemented CI/CD pipelines using Azure DevOps with automated testing, reducing manual deployment time by 60% and streamlining release workflows across development, staging, and production environments

Projects

SoundCanvas | *C++, Python, TensorFlow, Node.js, AWS, MySQL* | GitHub

- Designed an image-to-music system generating genre-conditioned audio with a 91% top-1 TensorFlow genre model
- Exposed a GraphQL API (Node.js) to orchestrate Dockerized C++/Python microservices on AWS ECS
- Provisioned AWS infrastructure with Terraform IaC for ECS Fargate, VPC/IAM, S3, and RDS MySQL

UTMIST Image Super-Resolution Pipeline | *Python, PyTorch, OpenCV, CUDA* | GitHub

- Implemented a PyTorch CNN with PixelShuffle and AMP for 480p-to-2K SR, cutting VRAM 40% on T4 GPUs
- Engineered a 2K SR data pipeline with dataset curation, a patch-based DataLoader, and OpenCV preprocessing
- Benchmarked the pipeline with PSNR/SSIM, delivering +3.1 dB PSNR and +0.05 SSIM vs. bicubic baseline

Publications

Second Author. **“Renewable Energy Integration in Northern Canada: A Machine Learning-Driven, Multi-Objective Evolutionary Framework.”** *Applied Energy*, awaiting publication.

Technical Skills

Languages: Python, Java, C/C++, C#, Swift, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks: ASP.NET Core, Entity Framework, Node.js, React, Flask, ROS
Libraries: PyTorch, TensorFlow, OpenCV, NumPy, SciPy, GraphQL
Developer Tools: AWS (S3, RDS, ECS), Terraform, Docker, Kubernetes, MySQL, SQL Server, Git, Linux, Azure DevOps